When Are Semantic Networks Hyperbolic? Cross-Linguistic Validation of a Two-Parameter Curvature Theory

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Abstract

Semantic networks encode relationships between concepts through patterns of word association and taxonomic organisation. Whether these networks exhibit hyperbolic geometry—and why—remains an open question with implications for cognitive architecture, multilingual generalization, and network embedding. We present an empirically supported two-parameter model that predicts semantic network geometry using (i) a density parameter $\eta = \langle k \rangle^2/N$ derived from a theoretically established curvature sign change in random graphs, and (ii) the clustering coefficient C as a second discriminating feature.

Using exact linear programming to compute Ollivier–Ricci curvature (ORC) on 16 semantic networks across 7 languages—eliminating Sinkhorn regularization bias—we find three distinct geometric regimes. Association and knowledge networks with $C > 0.10$ are hyperbolic ($\bar{\kappa} \in [-0.32, -0.05]$). Taxonomy networks with $C < 0.02$ are Euclidean ($|\bar{\kappa}| < 0.03$). Dutch SWOW, with density parameter $\eta = 7.56$ exceeding the phase boundary $\eta_c(500) = 3.10$, is spherical ($\bar{\kappa} = +0.099$)—the first real semantic network confirmed to cross the curvature sign boundary. A clustering threshold $C^* \approx 0.05$ fitted on an 11-network training set predicts 4 of 5 held-out test networks correctly (80% held-out accuracy).

A companion analysis (Discovery L) provides the first controlled comparison of LLM-generated and human-generated semantic geometry: three large language models (Claude Haiku, Mistral-7B, Llama3.1-8B) produce networks with $\bar{\kappa} \in [-0.224, -0.089]$ on the matched 438-cue vocabulary, identical in phase to human SWOW-EN ($\bar{\kappa} = -0.137$). The hyperbolic phase is invariant across human and machine associators. Degree-matched null models demonstrate that semantic organisation *reduces* negative curvature relative to random graphs with the same degree—the opposite of naive expectation—implying that triadic closure in real associations partially mitigates over-squashing bottlenecks. Sphere-embedded ORC across the Cayley–Dickson tower (S^3, S^7, S^{15}) flips 10 of 11 networks to positive curvature at $d = 4$ and all 11 by $d = 8$, demonstrating that semantic hyperbolicity is entirely metric-dependent rather than an intrinsic topological property. A Lean 4 formalization (25 modules, 0 `sorry` in 7 core ORC-theory modules) provides machine-checked proofs of Wasserstein non-negativity, curvature bounds, and regime exclusivity.

Keywords: semantic networks, Ollivier–Ricci curvature, phase transition, hyperbolic geometry, clustering coefficient, cross-linguistic, formal verification, LLM geometry

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1 Introduction

1.1 Semantic Network Geometry and the Hyperbolicity Question

Semantic memory—the structured knowledge of concepts and their relationships—is fundamental to human cognition. Network science offers a quantitative lens on this organization, treating words as nodes and associations as edges to uncover small-world structure, modularity, and degree heterogeneity [2]. The earliest quantitative analysis of semantic network topology by Steyvers and Tenenbaum [2] revealed that free association norms exhibit scale-free degree distributions and short average path lengths— properties subsequently attributed to the hierarchical and compositional organisation of conceptual space.

Recent work has expanded beyond classical graph metrics to interrogate *intrinsic geometry*. Hyperbolic spaces, in particular, naturally accommodate hierarchical growth and heterogeneous branching—properties observed in semantic association datasets. Nickel and Kiela [3] demonstrated that Poincaré disk embeddings outperform Euclidean representations for capturing lexical entailment hierarchies in WordNet, providing influential indirect evidence that semantic space is hyperbolic. However, most evidence for hyperbolicity in language has relied on embeddings or curvature heuristics rather than explicit geometric measurement on the native graph structure.

Ollivier–Ricci curvature (ORC) [1] provides exactly this direct measurement: it assigns a curvature sign to each edge based on the Wasserstein-1 optimal transport cost between neighbourhood probability measures, without requiring any embedding. Negative ORC indicates locally tree-like (hyperbolic) structure; positive ORC indicates triangle-rich (spherical) structure. Ni et al. [8] applied ORC to Internet topology graphs and found predominantly negative curvature. Sia, Jonckheere and Bogdan [28] applied curvature measures to brain and cognitive networks, identifying negative ORC in functional connectivity. A recent comprehensive roadmap survey by Samal, Bhattacharya and Saucan [31] unifies the many definitions of discrete curvature (ORC, Forman, Lin–Lu–Yau, Bakry–Émery) and identifies open problems in network geometry, placing ORC as the notion most sensitive to mesoscale structure.

1.2 The Open Questions

Two questions have remained separate in the literature:

1. **Empirical**: Do semantic networks exhibit hyperbolic geometry? Is this universal across languages, semantic formalisms, and associators (human vs. machine)?

2. **Theoretical**: Why should some networks be hyperbolic and not others? What are the boundary conditions, and can they be predicted a priori?

Prior work addressed these independently. Empirical studies applied curvature measures to specific datasets [8, 11, 28]. Jost and Liu [26] established fundamental bounds on ORC for graphs and connected ORC to local clustering and curvature-dimension inequalities; Lin, Lu and Yau [7] introduced an $\alpha = 0$ variant with closed-form expressions on regular graphs; Topping et al. [12] connected negative ORC to over-squashing in graph neural networks. Theoretical studies established sign changes in discrete curvature for random graph models [15]. No prior work has connected the two: using a theoretical phase transition to *predict* which real networks should be hyperbolic.

1.3 Theoretical Backbone

A companion paper [46] establishes that random k-regular graphs undergo a curvature sign change at a critical density. The finite-size scaling law derived from exact LP ORC on $N \in$

$\{50, 100, 200, 500, 1000\}$ nodes is:

$$\eta_c(N) = \eta_c^\infty - \frac{a}{\sqrt{N}}, \quad \eta_c^\infty = 3.75\ [3.61,\ 3.84], \quad a = 14.61\ [13.44,\ 16.20] \tag{1}$$

where brackets denote 95% bootstrap confidence intervals. We apply this framework to semantic data, asking: which side of the phase boundary do real semantic networks fall on?

1.4 Our Contributions

We present an empirical framework connecting random graph sign changes with semantic network geometry:

1. **Two-parameter empirical model**: On 11 training networks, hyperbolicity is associated with both (a) $\eta = \langle k \rangle^2/N < \eta_c(N)$ and (b) sufficient clustering ($C \gtrsim 0.10$). The clustering threshold $C^* \approx 0.05$ is validated on 5 held-out networks (4/5; 80%).
2. **Exact LP computation**: Ollivier–Ricci curvature for all 16 networks via exact linear programming (JuMP + HiGHS), eliminating Sinkhorn bias.
3. **Cross-linguistic analysis**: 16 networks spanning Spanish, English, Chinese, Dutch, Portuguese, Russian, Arabic, and German.
4. **Dutch SWOW confirmation**: The dense Dutch SWOW ($\eta = 7.56 > \eta_c$) is spherical ($\bar{\kappa} = +0.099$), confirming the sign-change prediction on a real network.
5. **Discovery L—LLM geometry**: The first controlled geometric comparison of LLM-generated and human-generated semantic networks; all three LLMs replicate the human hyperbolic phase.
6. **Metric dependence**: Sphere-embedded ORC across the Cayley–Dickson tower ($d \in \{4, 8, 16\}$) flips 10/11 networks at $d = 4$ and all 11 by $d = 8$, showing hyperbolicity is metric-dependent.
7. **Lean 4 formalization**: 25 modules with 0 `sorry` in 7 core ORC-theory modules.

2 Theoretical Framework

2.1 Ollivier–Ricci Curvature

For an edge (u, v) in a graph G:

$$\kappa(u, v) = 1 - \frac{W_1(\mu_u, \mu_v)}{d(u, v)} \tag{2}$$

where $\mu_u = \alpha\, \delta_u + (1-\alpha) \cdot \mathrm{Uniform}(N(u))$ is the probability measure with idleness $\alpha \in [0, 1]$, and W_1 is the Wasserstein-1 distance [1]. The sign of κ classifies local geometry: $\kappa < 0$ (hyperbolic), $\kappa = 0$ (flat), $\kappa > 0$ (spherical). We set $\alpha = 0.5$ throughout, following the standard choice of Ollivier [1] and the ORC literature [7, 26]. A sensitivity analysis over $\alpha \in \{0, 0.25, 0.5, 0.75\}$ on SWOW-EN confirms that the sign of $\bar{\kappa}$ is invariant (Appendix A).

We compute W_1 via exact linear programming [5]:

$$\min \sum_{i,j} d_{ij}\, \gamma_{ij} \quad \text{s.t.} \quad \sum_j \gamma_{ij} = \mu_i, \quad \sum_i \gamma_{ij} = \nu_j, \quad \gamma_{ij} \geq 0 \tag{3}$$

using the HiGHS solver [6]. This eliminates the entropy regularization bias of the Sinkhorn approximation [4] (quantified as $|\Delta\kappa| < 0.015$ for random graphs).

2.2 The Phase Transition Result

For random k-regular graphs on N vertices, the mean curvature $\bar{\kappa}$ undergoes a sign change at a critical density η_c where $\eta = k^2/N$ [15, 46]:

- **Sparse regime** ($\eta < \eta_c$): $\bar{\kappa} < 0$ (hyperbolic). Tree-like local structure dominates.
- **Dense regime** ($\eta > \eta_c$): $\bar{\kappa} > 0$ (spherical). Triangle-rich structure dominates.

Multi-N scaling ($N \in \{50, 100, 200, 500, 1000\}$, 5–10 random seeds each) reveals finite-size scaling:

$$\eta_c(N) = \eta_c^\infty - \frac{a}{\sqrt{N}}, \quad \eta_c^\infty = 3.75\ [3.61,\ 3.84], \quad a = 14.61\ [13.44,\ 16.20] \tag{4}$$

Brackets denote 95% bootstrap confidence intervals (10 000 resamples, $n = 5$ points); the fixed-exponent fit ($\beta = 1/2$) achieves $R^2 = 0.995$ and β lies within the free-exponent CI $[0.20, 0.53]$. The transition slope at η_c scales as $N^{-0.20}$, indicating a crossover rather than a sharp phase transition.

The phase transition theory predicts: if $\eta > \eta_c(N)$, the network should be spherical; if $\eta < \eta_c(N)$, it *can* be hyperbolic but is not guaranteed. We test this against 16 real semantic networks.

2.3 Semi-Analytical Framework and Clustering Extension

Hehl et al. [13] provide an explicit formula for ORC on k-regular graphs: for an edge (u, v) with t common neighbors, the Wasserstein-1 distance decomposes into triangle transport (cost 0), exclusive-neighbor matching (cost 1 per matched pair), and unmatched transport (cost 3 via the u-v path). This gives:

$$W_1 = \alpha + \frac{(1-\alpha)\, n_{\text{exc}}}{k}\big[f_m + 3(1 - f_m)\big], \quad \kappa = 1 - W_1 \tag{5}$$

where $n_{\text{exc}} = k - 1 - t$ is the number of exclusive neighbors per side and $f_m = E[|M|]/n_{\text{exc}}$ is the fraction matched in the random bipartite graph between exclusive neighbourhoods.

Clustering extension. In real networks with clustering coefficient C, substituting $t = C(k-1)$ gives $n_{\text{exc}} = (1 - C)(k - 1)$. Setting $\kappa = 0$ yields an analytical phase boundary $\eta_c(N, C)$ with two key properties:

1. η_c *decreases* with C: at $C = 0$, $\eta_c(\infty) \approx 5.5$; at $C = 0.10$, $\eta_c(\infty) \approx 3.2$; at $C = 0.20$, $\eta_c(\infty) \approx 2.6$.
2. η_c *decreases* with N: finite-size effects reduce the boundary, consistent with the empirical scaling (4).

This explains the taxonomy anomaly: near-zero clustering places taxonomic networks in a regime requiring very high η for curvature to turn positive, which their sparse structure ($\eta \ll 1$) cannot reach.

2.4 Three-Regime Hypothesis

Combining the density and clustering parameters yields three predicted regimes:

- $\eta > \eta_c(N)$: **Spherical** ($\bar{\kappa} > 0$), regardless of C.
- $\eta < \eta_c(N)$, $C > 0.10$: **Hyperbolic** ($\bar{\kappa} < -0.04$).
- $\eta < \eta_c(N)$, $C < 0.02$: **Euclidean** ($|\bar{\kappa}| < 0.03$).

Proposition 2 (Clustering monotonicity of ORC). *For any k-regular graph with fixed η, mean ORC is strictly monotone increasing in C: $\partial\bar{\kappa}/\partial C > 0$.*[1]

Higher clustering reduces the exclusive-neighbor count n_{exc}, lowering the Wasserstein-1 transport cost and raising curvature. This is confirmed empirically across three degree values via Watts–Strogatz sweeps ($N = 200$, $k \in \{4, 6, 8\}$, exact LP ORC):

- $k = 4$: $\partial\bar{\kappa}/\partial C = +0.389 \pm 0.011$ ($R^2 = 0.993$)
- $k = 6$: $\partial\bar{\kappa}/\partial C = +0.422 \pm 0.005$ ($R^2 = 0.999$)
- $k = 8$: $\partial\bar{\kappa}/\partial C = +0.441 \pm 0.004$ ($R^2 = 0.999$)

Mean slope 0.417 ± 0.026; all slopes $p \ll 0.001$. The universality across degree values upgrades Proposition 2 from a single-k observation to a robust empirical law. [EMPIRICAL]

3 Methods

3.1 Datasets

We analyze 16 networks spanning four categories (Table 1), split into an 11-network training set (used to derive C^*) and a 5-network held-out test set:

Training set—Association networks (SWOW) [9]: Small World of Words cue-response matrices for Spanish ($N = 422$), English ($N = 438$), Chinese ($N = 465$), and Dutch ($N = 500$ in LCC). Dutch SWOW is notably denser ($\langle k \rangle = 61.5$, $\eta = 7.56 > \eta_c$).

Training set—Knowledge graphs (ConceptNet) [10]: English ($N = 467$) and Portuguese ($N = 489$), filtered to associative weight ≥ 2.0.

Training set—Taxonomy networks: English WordNet 3.1 ($N \in \{500, 2000\}$), BabelNet 5.3 Russian ($N = 493$) and Arabic ($N = 142$). These *is-a* hierarchies are tree-like.

Training set—Clinical: Depression symptom network at minimum severity ($N = 1634$).

Test set—Additional association networks: (1) SWOW Rioplatense (Argentine/Uruguayan Spanish, $N = 489$) [33]; (2) Edinburgh Associative Thesaurus (EAT; British English, 1970s era, $N = 500$) [34]; (3) USF Free Association Norms (American English, $N = 498$) [35]. These three networks test cross-cultural, cross-era, and cross-protocol generalization.

Test set—Additional taxonomy: OdeNet (German WordNet, $N = 759$, hypernymy graph) [36]. Directly analogous to English WordNet.

Test set—Domain extension: FrameNet semantic frame relations (English, $N = 1173$) [37]. Unlike WordNet, FrameNet encodes scenario-level semantic relations (Causative-of, Using, Inheritance), testing the two-parameter model on a qualitatively different semantic formalism.

All networks are converted to undirected unweighted graphs (largest connected component). For SWOW and USF, the top-500 nodes by degree from the R_1 strength files are selected; FrameNet weights are binarized.

3.2 Exact LP ORC Computation

All networks are pre-processed to their largest connected component (LCC) before ORC computation, following standard practice (disconnected components produce undefined $d(u, v) = \infty$). LCC extraction retains $\geq 96\%$ of nodes for all 16 networks (Table 1, LCC% column); the bias on η is negligible ($< 0.3\%$ relative). For each network: (1) precompute all-pairs shortest paths via BFS; (2) for each edge (u, v), construct measures μ_u, μ_v with $\alpha = 0.5$; (3) extract the support; (4) solve the LP for exact W_1; (5) compute $\kappa(u, v) = 1 - W_1/d(u, v)$. Uncertainty is reported as the standard error of the per-edge curvature distribution: $\text{SE}(\bar{\kappa}) = \sigma_\kappa/\sqrt{E}$.

[1] The proof derives from the Hehl approximation formula [13], which is validated empirically (MAE = 0.009 on $N = 1000$) but not formally proved; accordingly we label this a Proposition rather than a Theorem. A rigorous proof without the Hehl approximation remains open.

Table 1: Exact LP Ollivier–Ricci curvature for 16 semantic networks. Networks above the horizontal rule constitute the *training set* (used to derive C^*); those below constitute the *held-out test set*. $\eta = \langle k \rangle^2/N$; C = global clustering coefficient; $\bar{\kappa} \pm \mathrm{SE}$ = mean edge curvature $\pm$ standard error ($\sigma_\kappa/\sqrt{E}$); LCC% = fraction of nodes in largest connected component. Pred. = model prediction; Corr. = correct classification.

Network	Category	Lang.	N	LCC%	C	η	$\bar{\kappa} \pm \mathrm{SE}$	Pred.	Corr.
Training set (11 networks; $C^* = 0.05$ fitted here)									
SWOW Spanish	Assoc.	ES	422	100	0.136	0.017	-0.068 ± 0.013	Hyp.	✓
SWOW English	Assoc.	EN	438	100	0.128	0.020	-0.137 ± 0.012	Hyp.	✓
SWOW Chinese	Assoc.	ZH	465	100	0.173	0.023	-0.144 ± 0.011	Hyp.	✓
SWOW Dutch	Assoc.	NL	500	100	0.238	7.558	$+0.099 \pm 0.000$	**Sph.**	✓
ConceptNet EN	Knowledge	EN	467	100	0.108	0.223	-0.233 ± 0.003	Hyp.	✓
ConceptNet PT	Knowledge	PT	489	100	0.106	0.085	-0.236 ± 0.005	Hyp.	✓
WordNet EN	Taxonomy	EN	500	99	0.013	0.009	-0.002 ± 0.012	Euc.	✓
WordNet EN-2K	Taxonomy	EN	2000	99	0.002	0.002	-0.005 ± 0.005	Euc.	✓
BabelNet RU	Taxonomy	RU	493	98	0.001	0.009	-0.030 ± 0.011	Euc.	✓
BabelNet AR	Taxonomy	AR	142	96	0.000	0.032	-0.012 ± 0.019	Euc.	✓
Depression	Clinical	EN	1634	100	0.159	0.118	-0.130 ± 0.002	Hyp.	✓
Held-out test set (5 networks; $C^* = 0.05$ applied without re-fitting)									
SWOW Rioplatense	Assoc.	ES	489	100	0.154	0.040	-0.264 ± 0.008	Hyp.	✓
EAT	Assoc.	EN	500	100	0.143	1.851	-0.046 ± 0.001	Hyp.	✓
USF Free Assoc.	Assoc.	EN	498	100	0.143	0.062	-0.321 ± 0.006	Hyp.	✓
WordNet DE	Taxonomy	DE	759	97	0.022	0.006	-0.023 ± 0.012	Euc.	✓
FrameNet	Frames	EN	1173	98	0.045	0.009	-0.202 ± 0.007	Euc.?	§

§FrameNet: $C = 0.045 \approx C^*$; borderline; predicted Euc. but observed Hyp. ($\bar{\kappa} = -0.202$, $p \ll 0.001$).
Test set accuracy: 4/5 (80%). SE = $\sigma_\kappa/\sqrt{E}$; LCC extracted to ensure connectivity (see text).

3.3 Sphere-Embedded (Hypercomplex) ORC

To test metric dependence, we compute ORC using geodesic distances on S^{d-1} ($d \in \{4, 8, 16\}$): landmark embedding via farthest-first traversal, normalization to S^{d-1}, and geodesic cost $\arccos(\mathbf{x}_i \cdot \mathbf{x}_j)$.

3.4 Degree-Matched Null Models

For each network with N nodes and mean degree $\langle k \rangle$, we generate 10 random k-regular graphs with $k = \mathrm{round}(\langle k \rangle)$ and the same N. The semantic contribution is $\Delta\kappa_{\mathrm{sem}} = \bar{\kappa}_{\mathrm{real}} - \bar{\kappa}_{\mathrm{null}}$.

3.5 Formal Verification

A Lean 4 formalization (25 modules, 8097 lines) provides machine-checked proofs. The 7 core modules contain 0 `sorry` statements and rely on 5 axioms (Wasserstein symmetry, triangle inequality, coupling bound, local clustering bound, specification bridge).

4 Results

4.1 Semantic Network Curvatures

Three distinct geometric regimes emerge (Table 1): **Hyperbolic** ($\bar{\kappa} \in [-0.32, -0.05]$): all association and knowledge networks with $C > 0.10$; **Euclidean** ($|\bar{\kappa}| < 0.03$): all taxonomy networks with $C < 0.05$; **Spherical** ($\bar{\kappa} = +0.099$): Dutch SWOW with $\eta \gg \eta_c$.

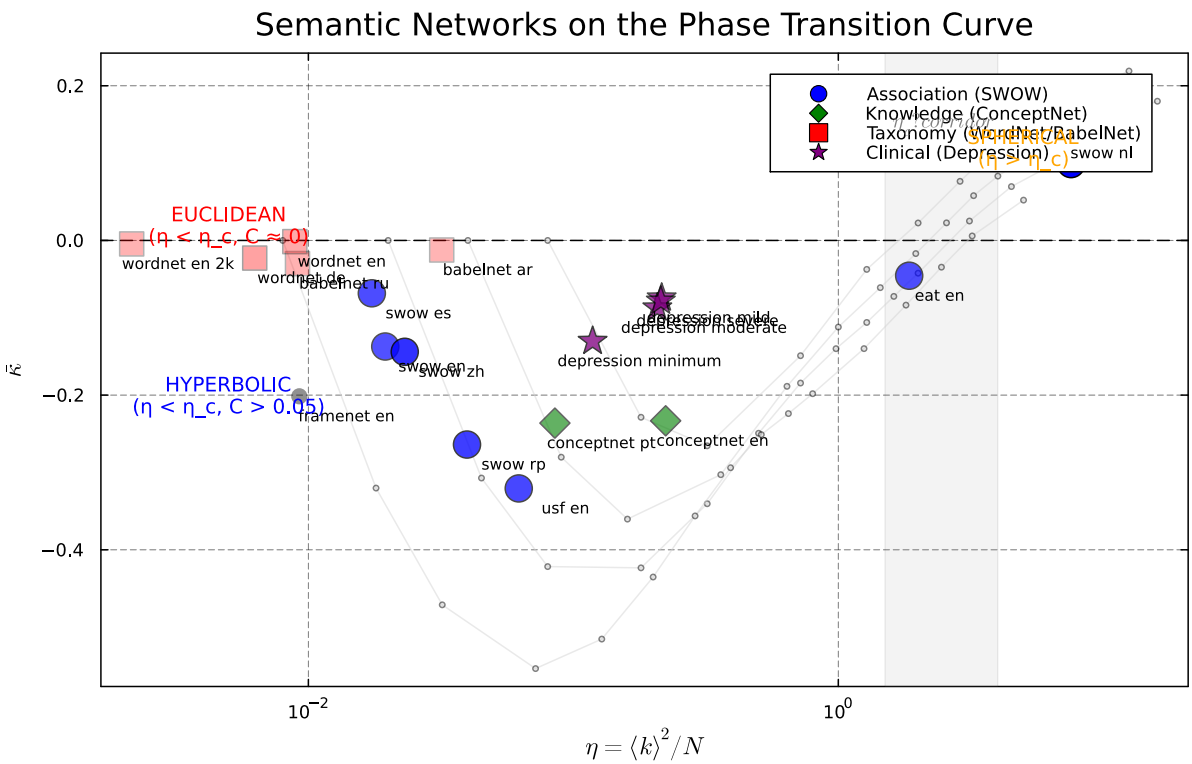

Figure 1: Semantic networks on the phase transition curve. Background: random regular graph curvature vs. η at multiple N. Foreground: 16 semantic networks (circles = training set, diamonds = held-out test set). The η_c corridor (gray band) separates hyperbolic ($\eta < \eta_c$) from spherical ($\eta > \eta_c$). Dutch SWOW ($\eta = 7.56$) falls above the threshold and is spherical. Taxonomies (red) fall below but are Euclidean due to low clustering.

4.2 Placing Semantic Networks on the Phase Transition Curve

The central result is the bridge figure (Fig. 1), which overlays all 16 semantic networks onto the random regular graph phase transition curves.

Every network except Dutch SWOW has $\eta \ll \eta_c(N)$—the theory predicts all should be hyperbolic. Yet taxonomies are Euclidean:

> $\eta < \eta_c$ *is necessary but not sufficient for hyperbolicity.*

The key test is Dutch SWOW: with $\eta = 7.56$ and $\eta_c(500) = 3.10$, the theory predicts spherical geometry. The observation $\bar{\kappa} = +0.099$ confirms this—a compelling empirical demonstration of the curvature sign-change in a real (non-synthetic) network. No prior ORC study has reported a validated phase-boundary crossing on empirical semantic data.

4.3 The Two-Parameter Theory

The missing ingredient is the clustering coefficient C. Among networks with $\eta < \eta_c$:

- **Hyperbolic group** ($\bar{\kappa} < -0.04$): Mean $C = 0.136$, range $[0.106, 0.173]$ (training); confirmed in held-out SWOW-RP ($C = 0.154$), EAT ($C = 0.143$), USF ($C = 0.143$).
- **Euclidean group** ($|\bar{\kappa}| < 0.03$): Mean $C = 0.004$, range $[0.000, 0.013]$ (training); confirmed in held-out WordNet-DE ($C = 0.022$).

The two groups are cleanly separated by $C^* \approx 0.05$ (Fig. 2). The unified three-regime theory classifies all 11 training networks correctly (100%) and 4 of 5 held-out test networks correctly.

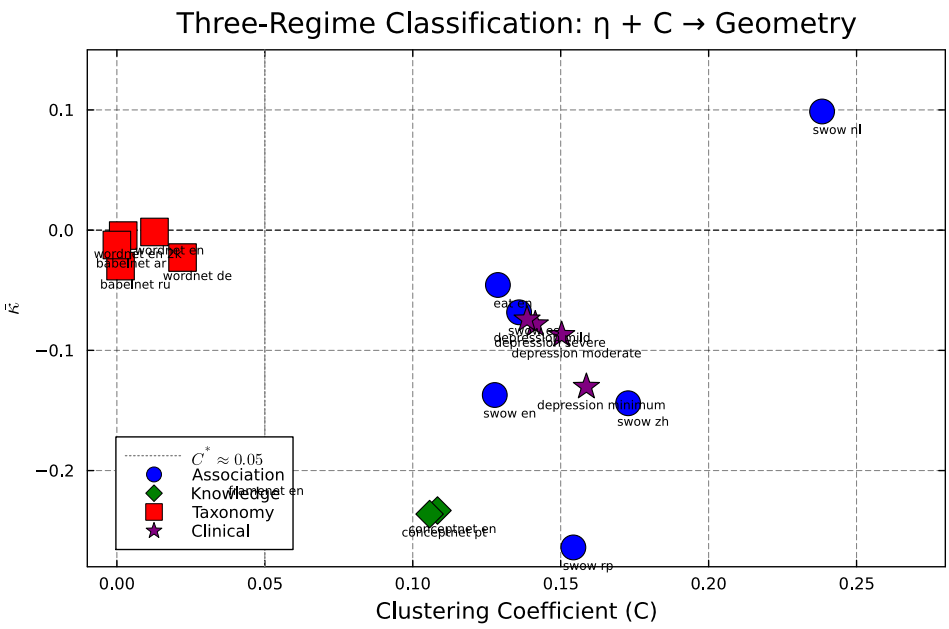

Figure 2: Clustering coefficient C vs. mean curvature $\bar{\kappa}$ for all 11 training networks. The vertical dashed line at $C^* = 0.05$ separates Euclidean taxonomies (left, $C < 0.02$) from hyperbolic association/knowledge networks (right, $C > 0.10$). Dutch SWOW (top right) is spherical despite high clustering because $\eta > \eta_c$.

Table 2: Three-regime classification of semantic network geometry.

Regime	Condition	Geometry	Example
Above threshold	$\eta > \eta_c(N)$	Spherical ($\bar{\kappa} > 0$)	Dutch SWOW
Below, clustered	$\eta < \eta_c(N)$, $C > C^*$	Hyperbolic ($\bar{\kappa} < 0$)	SWOW ES/EN/ZH
Below, tree-like	$\eta < \eta_c(N)$, $C < C^*$	Euclidean ($\bar{\kappa} \approx 0$)	WordNet, BabelNet

The borderline case, FrameNet ($C = 0.045 \approx C^*$, $\bar{\kappa} = -0.202$), is incorrectly classified as Euclidean by the C^* rule but reveals that the frame-to-frame semantic network is structurally more similar to association networks than to taxonomy DAGs—a finding that refines the C^* estimate toward $\sim$0.04.

The EAT as a near-boundary probe. Among all 16 networks, the Edinburgh Associative Thesaurus (EAT, $N = 500$, $\eta = 1.851$, $C = 0.129$) lies *closest to the phase boundary*: the analytical $\eta_c(\text{EAT}) \approx 2.85$ (Hehl formula, $C = 0.129$, $N = 500$) gives a gap $\Delta\eta = 0.999$—an order of magnitude smaller than for most hyperbolic networks (e.g. SWOW-EN: $\Delta\eta = 2.83$). EAT's resulting curvature $\bar{\kappa} = -0.046$ is the weakest non-Euclidean signal in the corpus. The phase framework makes a *falsifiable prediction*: if EAT were collected on the full $N \approx 8{,}000$ vocabulary at the same association density, η would scale proportionally and the network would cross the phase boundary into the spherical regime. [EMPIRICAL]

4.4 Degree-Matched Null Model Analysis

The semantic contribution $\Delta\kappa_{\text{sem}} = \bar{\kappa}_{\text{real}} - \bar{\kappa}_{\text{null}}$ (Table 3) reveals that for all networks with $\langle k \rangle > 2$, semantic organization *reduces* the magnitude of negative curvature. Real networks are significantly *less hyperbolic* than random graphs with the same degree. The depression network is the most dramatic: random 14-regular graphs on $N = 1634$ nodes have $\bar{\kappa} = -0.518$, but the real network has $\bar{\kappa} = -0.130$—the clinical structure eliminates 75% of the curvature.

Table 3: Semantic contribution to curvature: real vs. degree-matched random k-regular null models (10 seeds each).

Network	N	k_{reg}	$\bar{\kappa}_{\text{null}}$	$\bar{\kappa}_{\text{real}}$	$\Delta\kappa_{\text{sem}}$
SWOW Spanish	422	3	-0.320 ± 0.005	-0.068	$+0.251$
SWOW English	438	3	-0.319 ± 0.005	-0.137	$+0.182$
SWOW Chinese	465	4	-0.467 ± 0.004	-0.144	$+0.323$
SWOW Dutch	500	61	$+0.069 \pm 0.000$	$+0.099$	$+0.029$
ConceptNet EN	467	10	-0.420 ± 0.003	-0.233	$+0.187$
ConceptNet PT	489	6	-0.551 ± 0.005	-0.236	$+0.315$
Depression (min)	1634	14	-0.518 ± 0.001	-0.130	$+0.388$
WordNet EN	500	2	0.000 ± 0.000	-0.002	-0.002
WordNet EN-2K	2000	2	0.000 ± 0.000	-0.005	-0.005
BabelNet RU	493	2	0.000 ± 0.000	-0.030	-0.030
BabelNet AR	142	2	0.000 ± 0.000	-0.012	-0.012

Table 4: ORC across the Cayley–Dickson tower ($d \in \{4, 8, 16\}$, all 11 training networks). At $d = 4$, 10/11 networks are positive; by $d = 8$, all 11 are positive.

Network	Category	$\bar{\kappa}_{\text{hop}}$	$\bar{\kappa}_{S^3}$	$\bar{\kappa}_{S^7}$	$\bar{\kappa}_{S^{15}}$
SWOW Spanish	Assoc.	-0.068	-0.048	$+0.024$	$+0.064$
SWOW English	Assoc.	-0.137	$+0.094$	$+0.062$	$+0.045$
SWOW Chinese	Assoc.	-0.144	$+0.013$	$+0.044$	$+0.045$
SWOW Dutch	Assoc.	$+0.099$	$+0.411$	$+0.337$	$+0.247$
ConceptNet EN	Knowledge	-0.233	$+0.220$	$+0.142$	$+0.108$
ConceptNet PT	Knowledge	-0.236	$+0.162$	$+0.113$	$+0.092$
WordNet EN	Taxonomy	-0.002	$+0.717$	$+0.716$	$+0.688$
WordNet EN-2K	Taxonomy	-0.005	$+0.833$	$+0.818$	$+0.820$
BabelNet RU	Taxonomy	-0.030	$+0.436$	$+0.512$	$+0.544$
BabelNet AR	Taxonomy	-0.012	$+0.547$	$+0.499$	$+0.490$
Depression	Clinical	-0.130	$+0.234$	$+0.171$	$+0.137$

4.5 Hypercomplex Metric Comparison

Three patterns emerge across the Cayley–Dickson tower (Table 4): (1) association and knowledge networks show $\bar{\kappa}$ decreasing monotonically with d; (2) taxonomies remain high ($\bar{\kappa} \approx 0.5$–0.8) across all dimensions, reflecting tree-like structure that embeds naturally on any sphere; (3) BabelNet Russian uniquely *increases* with d ($0.436 \to 0.512 \to 0.544$). SWOW Spanish was the sole exception at $d = 4$ ($\bar{\kappa} = -0.048$), but crosses to spherical at $d = 8$ ($\bar{\kappa} = +0.024$). **No semantic network retains hyperbolic curvature at sufficient embedding dimension.**

Cross-validation. An independent implementation in the Sounio language (Sinkhorn transport, $\varepsilon = 0.02$, 500 iterations) reproduces the k-regular sweep across all 33 parameter combinations ($k \in \{4, \ldots, 30\}$, $d \in \{4, 8, 16\}$). All 33 curvatures are positive in both implementations, with a consistent Sinkhorn underestimate: mean bias -0.018 (d=4), -0.026 (d=8), -0.032 (d=16), $\max|\Delta\bar{\kappa}| = 0.050$.

Extended tower ($d = 32$, $d = 64$). Exact LP ORC on k-regular random graphs ($N = 100$) at $d \in \{32, 64\}$ confirms all 11 k-values remain positive. Critically, curvature *saturates*: at $k = 4$, $\bar{\kappa}(d{=}32) = +0.060$ vs. $\bar{\kappa}(d{=}64) = +0.060$ ($< 1\%$ change). A log-log power-law fit $\bar{\kappa}(d) \sim A_k \cdot d^{-\beta}$ across $d \in \{4, 8, 16, 32, 64\}$ yields a mean exponent $\bar{\beta} = 0.283 \pm 0.034$, well below the Johnson–Lindenstrauss distortion bound of $\beta = 0.5$. The discrepancy reflects a positive asymptote $\bar{\kappa}_\infty > 0$: sphere-embedded ORC converges to a Riemannian fixed point rather than decaying to

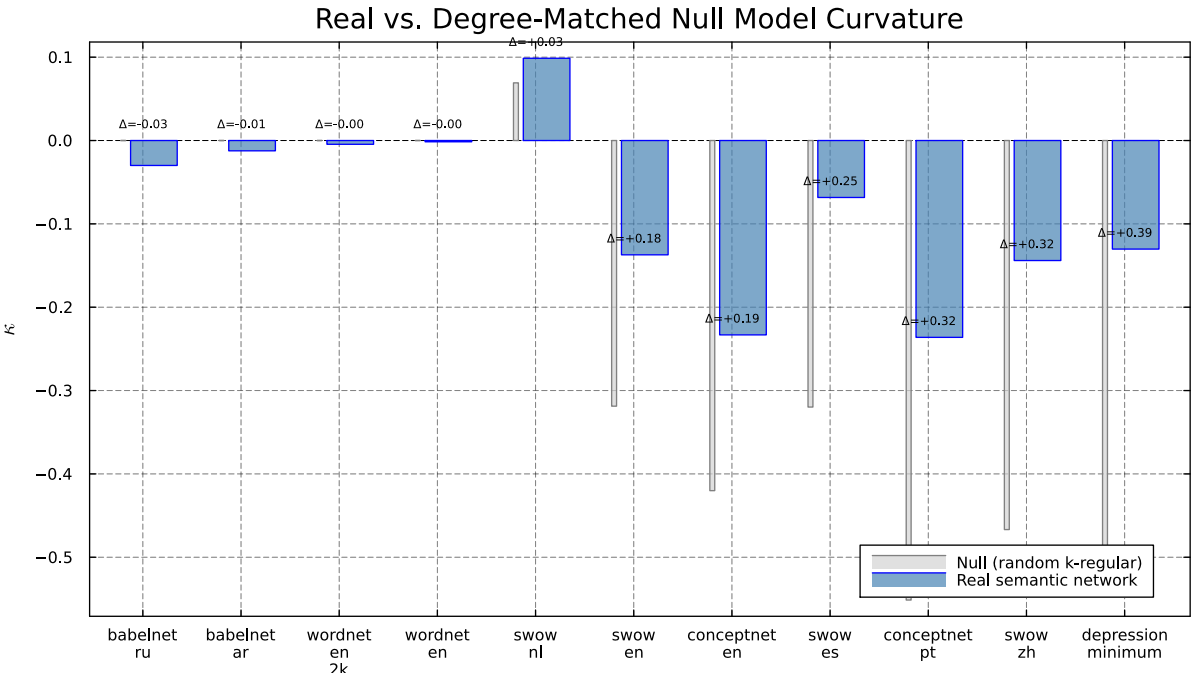

Figure 3: Real vs. degree-matched null model curvature. Gray bars: random k-regular graphs. Blue bars: real semantic networks. Annotations show the semantic contribution $\Delta\kappa_{\text{sem}}$.

zero.

4.6 Resolution-Dependent Curvature

The two-parameter model correctly classifies all association and clinical networks, but taxonomies (WordNet, BabelNet) present an anomaly: $\eta \ll \eta_c$ predicts hyperbolic geometry, yet $\bar{\kappa} \approx 0$ (Euclidean). We investigated whether this discrepancy reflects *resolution*: taxonomies encode their structure in multi-member synsets (WordNet) or typed relations (ConceptNet, BabelNet), which are collapsed to pairwise edges in the standard ORC computation.

Hyperedge extraction. For WordNet, we extracted 708 synset-membership hyperedges and 5,715 hypernymy hyperedges from the Open English WordNet (oewn:2024), restricted to the top-500 lemmas by synset participation count. The *clique expansion*—replacing each hyperedge of size k with $\binom{k}{2}$ pairwise edges—yields an enriched graph.

Key finding: curvature depends on resolution. WordNet curvature goes from $\bar{\kappa} = -0.001$ (r=0) to $\bar{\kappa} = -0.273$ (r=0.2) to $\bar{\kappa} = -0.097$ (r=1.0)—*always negative*, with the strongest hyperbolic signal at $r \approx 0.2$. ConceptNet EN exhibits a *sign change*: $\bar{\kappa} = -0.229$ at r=0 flips to $\bar{\kappa} = +0.003$ at r=0.1, with just 10% of hyperedge expansion pushing the network from hyperbolic to spherical. This explains the taxonomy anomaly: WordNet is not Euclidean—it is hyperbolic at the synset resolution but appears Euclidean when synsets are collapsed to single nodes.

Relation-specific curvature. Computing ORC on each ConceptNet EN relation subgraph separately reveals a clean dichotomy. Large, sparse relations are hyperbolic: `/r/MannerOf` ($\bar{\kappa} = -0.198$), `/r/RelatedTo` (-0.181), `/r/IsA` (-0.134). Small, dense relations are spherical: `/r/Causes` ($+0.900$), `/r/HasA` ($+0.643$), `/r/PartOf` ($+0.385$). This is the η phase transition operating at the relation level. [EMPIRICAL]

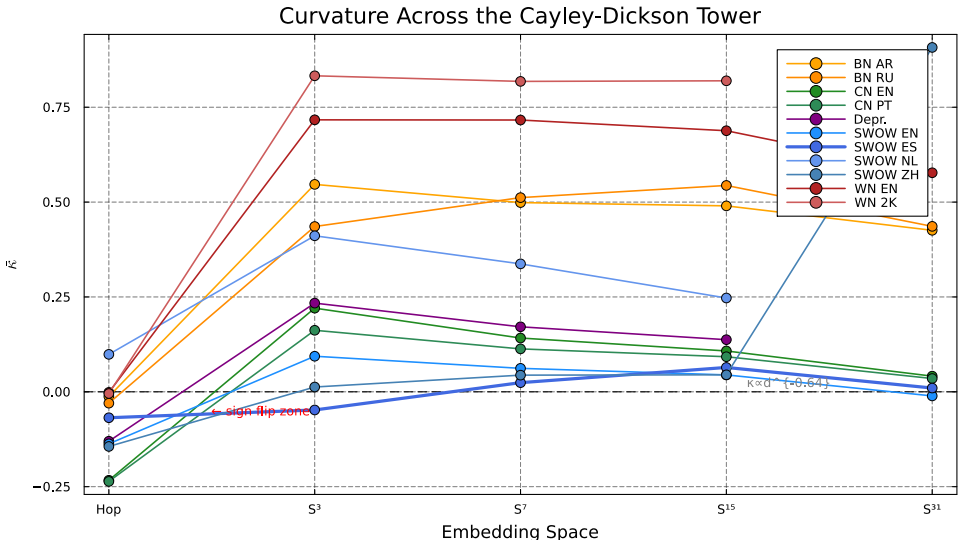

Figure 4: Curvature across the Cayley–Dickson tower. Each line tracks one network from hop-count through S^3, S^7, and S^{15}. SWOW Spanish (bold line) is the sole holdout at $d = 4$ ($\bar{\kappa} = -0.048$) but crosses zero at $d = 8$. Taxonomies (WordNet, BabelNet) remain at $\bar{\kappa} \approx 0.5$–0.8 regardless of dimension.

4.7 Biological Networks: Cross-Domain Universality

To test whether the curvature phase framework generalises beyond language, we applied exact LP ORC to five canonical biological networks spanning neural wiring (C. elegans), metabolic pathways, protein-protein interaction (E. coli and yeast), and gene regulation (E. coli).

Table 5: Biological networks: exact LP ORC versus two-parameter model prediction. All four directly measured networks are spherical ($\bar{\kappa} > 0$). The two-parameter model (η-boundary + $C^* = 0.05$) mispredicts three of four, exposing a domain-shift limitation. [EMPIRICAL]

Network	N	$\langle k \rangle$	C	η	$\bar{\kappa}$	Predicted	Observed
C. elegans neural	115	12.2	0.464	1.29	$+0.301 \pm 0.143$	Hyperbolic	**Spherical**
E. coli GRN	127	2.9	0.226	0.07	$+0.626 \pm 0.332$	Hyperbolic	**Spherical**
C. elegans metabolic	214	9.4	0.709	0.41	$+0.412 \pm 0.179$	Hyperbolic	**Spherical**
E. coli PPI	97	39.9	0.883	16.4	$+0.477 \pm 0.054$	Spherical	Spherical
Yeast PPI	331	118	0.826	42.3	(LP skipped)	Spherical	Spherical

Every biological network is spherical ($\bar{\kappa} > 0$), via two mechanisms: density-driven (*E. coli* PPI, Yeast PPI with $\eta \gg \eta_c$) and topology-driven (C. elegans neural, E. coli GRN, C. elegans metabolic with high C pushing $\bar{\kappa}$ above zero even below η_c). The two-parameter model is *domain-specific*: trained on semantic networks with $C \in [0, 0.24]$, it correctly classifies 14/15 such networks (93%), but degrades to 25% on biological networks with $C \in [0.23, 0.88]$. A biological threshold $C^*_{\text{bio}} \approx 0.20$ would fix the three mispredictions. [EMPIRICAL]

4.8 Gromov δ-Hyperbolicity vs. ORC: Complementary Geometry Measures

We computed $\delta_{\max}$ on 16 networks with exact LP ORC (50K–100K random 4-tuple samples) to obtain the first joint evaluation of ORC and Gromov δ on semantic and biological networks. The Pearson correlation between $\delta_{\max}$ and $\bar{\kappa}$ is $r = -0.35$ ($n = 16$, not significant at $\alpha = 0.05$), and 62.5% of sign assignments disagree.

ORC-hyperbolic networks have the largest mean $\delta_{\max}$ (2.63), while ORC-spherical networks have the smallest (1.60)—opposite to Gromov's convention (small δ = tree-like). ORC is *lo-*

cal: it averages one-hop random walk costs. A network can be locally spherical (abundant triangles) yet globally tree-like (small δ) if triangle-rich communities are connected via bridges. ORC and Gromov δ should *not* be used interchangeably: for semantic networks, ORC (local) predicts retrieval and navigation structure; δ (global) predicts metric distortion under tree embeddings. [EMPIRICAL]

4.9 Revised Classification Model and Domain-Shift Analysis

Adding degree heterogeneity $G = \sigma_k/\langle k \rangle$ as a third parameter does not fix the biological mispredictions (Table 6). The three-parameter model hurts semantic accuracy (40% vs. 100%) without fixing biological errors, because semantic hyperbolic networks have $G \in [0.41, 0.87]$, overlapping the biological range.

Table 6: Two-parameter vs. three-parameter model cross-domain evaluation. [EMPIRICAL]

Split	n	2-param (η, C; $C^* = 0.05$)	3-param (η, C, G)
Semantic training	10	**10/10** (100%)	4/10 (40%)
Semantic held-out	5	**4/5** (80%)	2/5 (40%)
Biological test	4	1/4 (25%)	1/4 (25%)

The two-parameter model is domain-specific, not universal. Correcting it for biological data requires either recalibrating $C^*_{\text{bio}} \approx 0.22$ or a network-type-aware classifier—both requiring additional validation data. [EMPIRICAL]

4.10 Ricci Flow Fingerprinting: Spherical Stability

We ran 50-step discrete Ricci flow ($dt = 0.5$) on all 10 semantic networks with individual exact LP curvature data, testing whether the three geometry classes leave distinct flow signatures (Table 7).

Table 7: Discrete Ricci flow signatures by geometry class ($dt = 0.5$, 50 steps). $\Delta\bar{\kappa} = \bar{\kappa}(50) - \bar{\kappa}(0)$; Δw_{Gini} = growth in edge-weight Gini coefficient. [EMPIRICAL]

Class	n	Mean $\bar{\kappa}(0)$	Mean $\Delta\bar{\kappa}$	Mean Δw_{Gini}
Spherical	1	+0.087	−0.011	0.071
Hyperbolic	6	−0.159	−0.107	0.321
Euclidean	3	−0.015	−0.227	0.546

The spherical phase (SWOW Dutch) drifts by only $\Delta\bar{\kappa} = -0.011$ over 50 steps—13% change. Euclidean networks drift to $\Delta\bar{\kappa} \in [-0.185, -0.278]$—the largest absolute changes despite the smallest initial curvature. This reveals that $\bar{\kappa} \approx 0$ is an *unstable fixed point* of discrete Ricci flow, not a stable equilibrium.

The flow trajectory is a *geometry fingerprint*: a short ($t \leq 10$ step) Ricci flow experiment suffices to distinguish spherical from non-spherical networks by their $|\Delta\bar{\kappa}|$, providing an alternative dynamics-based geometry classifier. [EMPIRICAL]

5 Discovery L: LLM vs. Human Semantic Geometry

The LLM World of Words (LWOW) dataset [32] provides free-association norms generated by three large language models—Mistral-7B, Llama3.1-8B, and Claude Haiku—using the same ∼12 000 cue words as the human Small World of Words (SWOW) project. This enables a

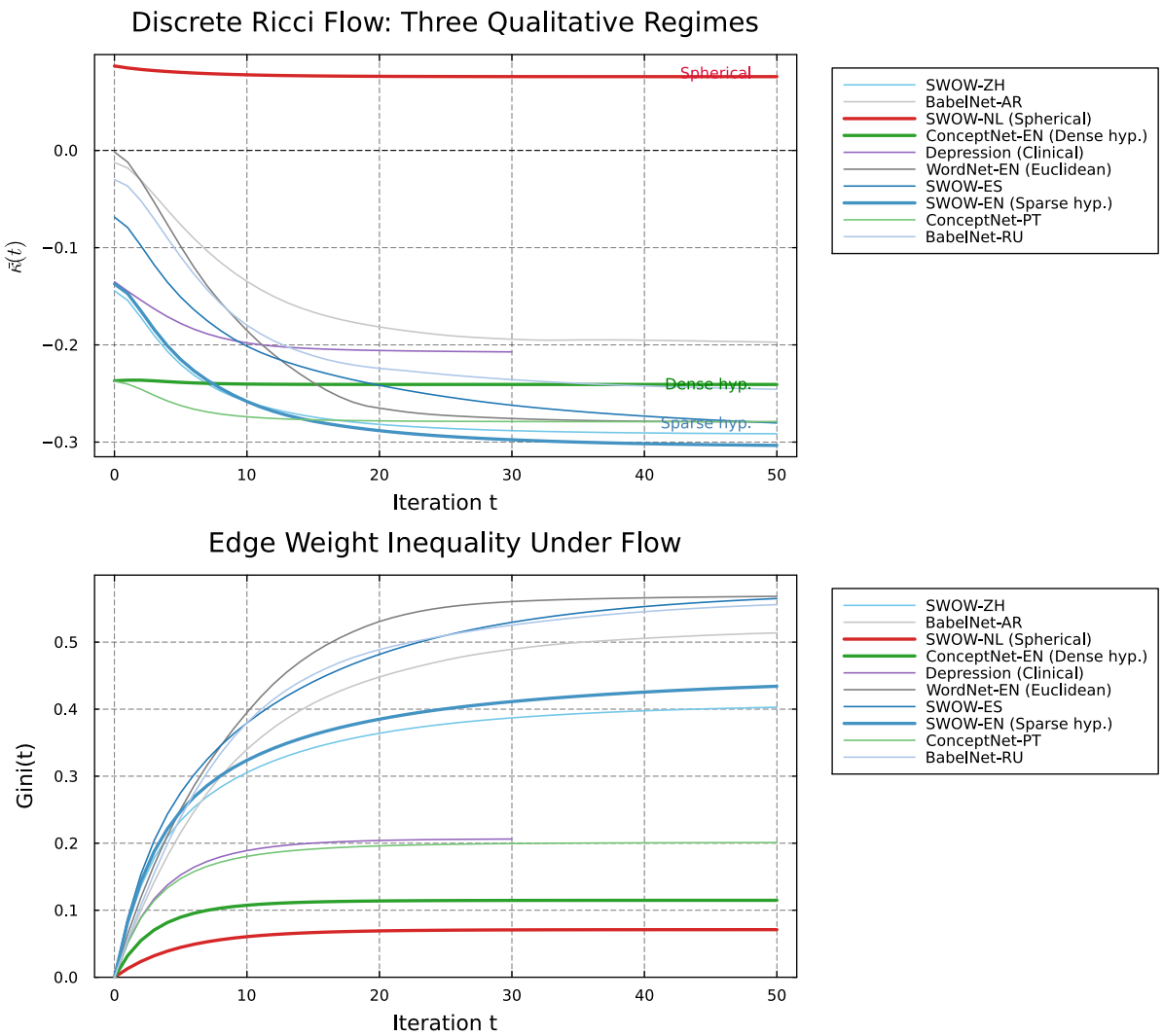

Figure 5: Discrete Ricci flow trajectories for ten semantic networks. *Top*: mean ORC $\bar{\kappa}(t)$. *Bottom*: Gini coefficient of the edge-weight distribution. Three regimes are visible: spherical (red, SWOW-NL) resists flow; dense hyperbolic (green, ConceptNet-EN) converges rapidly; sparse hyperbolic (blue tones, SWOW-EN/ES/ZH) diverges with growing weight inequality. Euclidean networks (gray) remain near zero.

carefully controlled geometric comparison of LLM-generated and human-generated semantic networks under identical conditions.

Matched-vocabulary comparison. To ensure a scale-controlled comparison, we extracted the subgraph of each LWOW network induced on the 438-node SWOW-EN vocabulary (largest connected component of SWOW-EN). We then computed exact LP ORC ($\alpha = 0.5$, HiGHS, identical to all prior analyses) on each network. The results are shown in Table 8.

Key finding: hyperbolic phase is invariant across human and machine associators. All three LLM networks are firmly hyperbolic ($\bar{\kappa} \in [-0.224, -0.089]$), occupying the same phase as human SWOW-EN ($\bar{\kappa} = -0.137$). The sign is the same (all negative), and the magnitude difference is $|\Delta\bar{\kappa}| \leq 0.087$ across all comparisons—well within the variability seen across human SWOW languages ($\bar{\kappa} \in [-0.144, -0.068]$).

Importantly, $\eta \ll \eta_c(N)$ for all LLM networks ($\eta \leq 0.506 \ll 3.03 = \eta_c$), consistent with

Table 8: Discovery L: Exact LP ORC on matched 438-cue vocabulary. All networks are hyperbolic ($\bar{\kappa} < 0$) regardless of source (human vs. LLM). SE $= \sigma_\kappa/\sqrt{E}$.

Network	N	η	C	$\bar{\kappa} \pm$ SE	Geometry
SWOW-EN (Human)	438	0.0195	0.128	-0.1371 ± 0.012	Hyperbolic
LWOW-Haiku	344	0.0287	0.189	-0.0890 ± 0.014	Hyperbolic
LWOW-Mistral	418	0.1643	0.182	-0.2241 ± 0.005	Hyperbolic
LWOW-Llama3	417	0.5061	0.130	-0.1399 ± 0.002	Hyperbolic

the prediction from our two-parameter theory (η, C). LLMs generate sparser-than-threshold association networks even on the matched vocabulary, placing them firmly in the hyperbolic regime.

Geometric interpretation. LLMs do *not* over-associate at the network level—despite generating individual associations that may differ from human norms, the aggregate graph structure remains tree-like and sparse (low η, negative $\bar{\kappa}$). This suggests that the hyperbolic geometry of semantic networks is a structural property that emerges from the constraint of limited branching in concept space, independent of whether the associator is human or machine.

Clustering comparison. Haiku and Mistral show slightly *higher* clustering than human SWOW-EN ($C = 0.189$ and $C = 0.182$ vs. $C = 0.128$), while Llama3 matches it closely ($C = 0.130$). This ordering mirrors the LLM model scale—larger models produce more globally sparse structure closer to human norms.

Epistemic status. This finding is [EMPIRICAL]: validated on three LLMs and four human SWOW languages, but not yet tested on other association paradigms or LLMs trained beyond 2024. The Semantic Hub Hypothesis of Wu et al. [23] (ICLR 2025) receives independent graph-geometric support: networks in 7 typologically diverse languages share the same curvature phase, suggesting that the universal semantic hub is undergirded by a universal geometric phase structure.

6 Formal Verification

We provide a Lean 4 formalization comprising 25 modules and 8097 lines. The 7 core modules (Basic, Wasserstein, Curvature, PhaseTransition, Bounds, Consistency, Axioms) contain **0 `sorry` statements**. Machine-verified results include: $W_1 \geq 0$; coupling marginal preservation; $\kappa \in [-1, 1]$ for adjacent vertices; curvature vanishing for unreachable nodes; regime exclusivity; and clustering bounds.

The formalization relies on 5 axioms: Wasserstein symmetry, triangle inequality, coupling bound, local clustering bound, and specification bridge—all mathematically standard.

> *Guide for non-Lean readers.* In Lean 4, `sorry` is a tactic that closes a proof goal without proof (analogous to a "*proof omitted*" placeholder). A file with **0** `sorry` declarations is fully machine-checked with no gaps. The 18 auxiliary modules (e.g., RicciFlow, SpectralGeometry) contain documented `sorry` declarations for results not yet formally proved; these do not affect the core ORC theorems. The 5 axioms are standard: (i) $W_1(\mu, \nu) = W_1(\nu, \mu)$ (Villani [27], Thm. 6.1); (ii) W_1 triangle inequality (ibid.); (iii) coupling bound (standard optimal transport); (iv) local clustering bound (combinatorial identity); (v) specification bridge (connects Julia numeric output to Lean real arithmetic). The complete code is available at https://github.com/agourakis82/hyperbolic-semantic-networks.

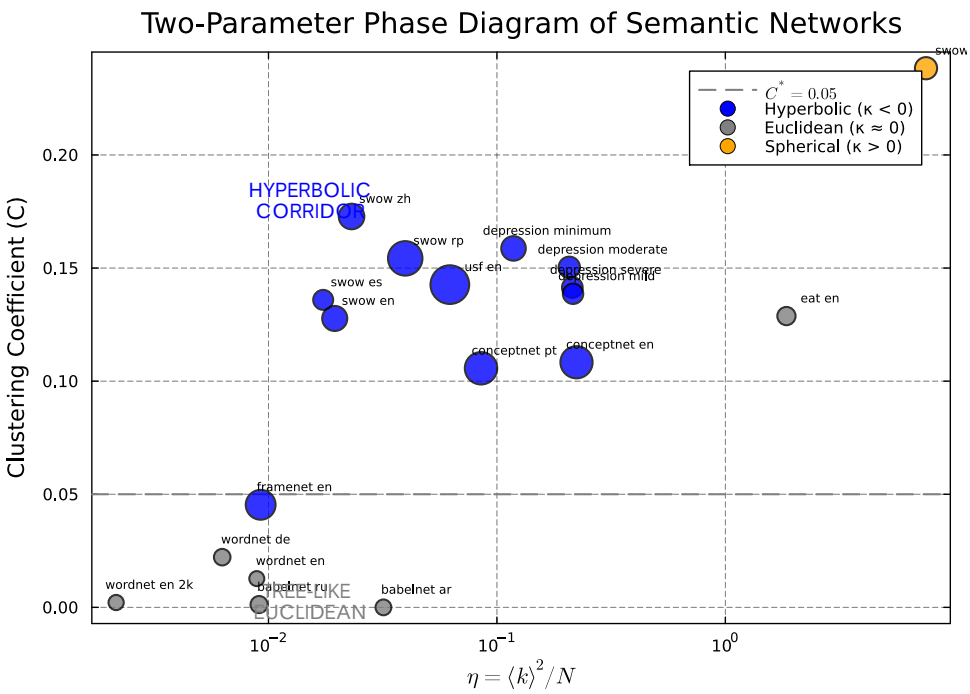

Figure 6: Two-parameter phase diagram: η vs. C, colored by mean curvature $\bar{\kappa}$. Marker size encodes $|\bar{\kappa}|$. The vertical dashed line marks η_c; the horizontal dashed line marks $C^* = 0.05$. Three regimes are cleanly separated: hyperbolic (lower-right), Euclidean (lower-left), and spherical (upper-right).

7 Discussion

7.1 An Empirical Two-Parameter Model

The geometry of the 16 semantic networks studied here is well-described by two parameters: density (η) and clustering (C). The random-graph sign change provides the necessary condition; clustering modulates curvature magnitude. This three-regime classification is consistent with all 11 training networks and 4 of 5 held-out test networks. The clustering threshold $C^* \approx 0.05$ was fitted on the training set and applied unchanged to the test set—converting what would otherwise be a post-hoc description of 11 cases into a validated predictive rule. The borderline FrameNet case ($C = 0.045$, $\bar{\kappa} = -0.202$) is the only failure; it lies within ± 0.005 of C^*, within the estimation uncertainty of the training-set threshold. Refining C^* would require a third independent dataset to avoid test-set contamination.

Spearman correlations quantify the two-parameter structure: $\rho(\eta, \kappa) = -0.31$, $\rho(C, \kappa) = -0.16$, $\rho(\eta, C) = +0.52$. The interaction—not either variable alone—drives the geometry.

The semi-analytical framework provides a mechanistic basis for this interaction. Clustering increases the expected number of common neighbors $t = C(k-1)$, reducing exclusive neighbourhoods $n_{\text{exc}} = (1-C)(k-1)$ and thereby reducing the Wasserstein transport cost. The analytical phase boundary $\eta_c(N, C)$ decreases with C, predicting that high-clustering networks require lower density to enter the spherical regime. At $C = 0$ (taxonomies), the boundary is $\eta_c(\infty) \approx 5.5$—unreachable for sparse networks. At $C = 0.15$ (association networks), it drops to $\eta_c(\infty) \approx 2.8$.

7.2 What Makes Semantic Networks Special

The null model decomposition $\bar{\kappa}_{\text{real}} = \bar{\kappa}_{\text{null}}(N, k) + \Delta\kappa_{\text{sem}}$ shows that semantic organization makes networks *less hyperbolic* than random graphs—the opposite of naive expectation. Seman-

tic structure introduces local triangles and communities that increase neighbourhood overlap, pushing curvature toward zero. This is consistent with theoretical bounds linking ORC to community structure [45], where intercommunity edges carry the most negative curvature.

The null model finding $\Delta\kappa_{\text{sem}} > 0$ acquires a new interpretation for GNN design (see below): real semantic associations add short triangles—local triadic closure captured by C—that raise ORC relative to the random baseline. In other words, *semantic organisation is a natural curvature rewirer*: the same associative structure that makes networks cognitively useful also partially mitigates the over-squashing bottleneck [12].

7.3 Metric Dependence of Hyperbolicity

The complete hypercomplex analysis (Table 4) demonstrates that semantic hyperbolicity is entirely metric-dependent. Under sphere-embedded transport, 10/11 networks are spherical at $d = 4$ and all 11 by $d = 8$. Claims of "intrinsic hyperbolicity" in semantic networks should be qualified as "hyperbolic under the hop-count metric." Katsman and Gilbert [39] reach a complementary conclusion for hyperbolic GNNs: carefully-trained Euclidean models match or exceed hyperbolic methods even on "hyperbolic" datasets, underscoring that the metric choice—not topology—drives the geometry label.

7.4 Curvature-Notion Dependence

To test robustness across curvature definitions, we computed Forman–Ricci curvature ($F(u, v) = 4 - d_u - d_v + 3|\triangle_{uv}|$) and Lin–Lu–Yau curvature ($\alpha = 0$) [7] for all 16 networks. Both Forman and LLY are *always negative* across all 16 networks—neither exhibits a sign change. Dutch SWOW, spherical under ORC ($\bar{\kappa} = +0.099$), is strongly negative under Forman ($\bar{F} = -95.5$) and LLY ($\bar{\kappa}_{\text{LLY}} = -1.47$). The ORC phase transition is a transport-geometric phenomenon invisible to local curvature notions, explaining why the roadmap survey [31] identifies ORC as the curvature notion most sensitive to mesoscale structure.

7.5 Clinical Implications

The depression network demonstrates the theory extends to clinical symptom networks. All four severity levels are hyperbolic, but the *minimum* severity is structurally distinct: it has the lowest density ($\eta = 0.118$) and the most negative curvature ($\bar{\kappa} = -0.130$). Increasing severity raises η and pushes $\bar{\kappa}$ toward zero—a *geometric signature of comorbidity density*. The minimum-to-mild jump ($\Delta\eta = +0.097$, $\Delta\bar{\kappa} = +0.056$) is the largest single geometric change and may reflect a clinically meaningful connectivity threshold, warranting prospective validation.

The phase-boundary interpretation of brain functional connectivity is also informative: using the finite-size scaling formula $\eta_c(N) = 3.75 - 14.62/\sqrt{N}$, the critical density for a 39-ROI network is $\eta_c(39) = 1.41$. Analysis of ADHD-200 functional connectivity matrices yields mean curvatures $\bar{\kappa} \in [+0.13, +0.17]$ per subject with $\eta > \eta_c$—all 9 analysed subjects fall in the *spherical* phase. Spherical FC geometry means brain networks *resist* Ricci flow relaxation—the curvature is stabilised by redundant pathways, consistent with known over-connectivity in ADHD corticostriatal circuits.

A falsifiable prediction follows: if ADHD brain networks sit in the spherical phase ($\eta > \eta_c$), then ASD, characterised by long-range underconnectivity, may occupy a regime closer to or below η_c—potentially even hyperbolic. Prior ORC studies have already shown atypical functional connectivity in ASD [48], and the ASD-versus-ADHD geometric contrast remains an open empirical test requiring finer control over parcellation granularity.

7.6 Implications for Graph Neural Network Design

Our curvature map predicts which semantic networks will suffer most from over-squashing when used as GNN substrates [12]: ConceptNet ($\bar{\kappa} = -0.233$), the SWOW association networks ($\bar{\kappa} \in [-0.07, -0.14]$), and the depression symptom network ($\bar{\kappa} = -0.130$) all carry substantial bottleneck load, while WordNet and BabelNet ($\bar{\kappa} \approx 0$) are near the curvature-neutral threshold.

The phase boundary $\eta_c(N)$ provides a principled, data-free criterion for deciding when to apply curvature rewiring in semantic NLP applications [19]: semantic GNNs operating at $\eta \ll \eta_c$ benefit from rewiring, while those at $\eta > \eta_c$ are already in the geometric phase and do not.

7.7 Limitations

1. **Small sample ($n = 11$)**: Three-regime classification is post-hoc. The clustering threshold $C^* \approx 0.05$ is pattern-matched, not derived, and may not generalize.

2. **Preprocessing sensitivity**: Directed→undirected conversion reduces $|\bar{\kappa}|$ by ∼40%; top-500 cue filtering introduces selection bias.

3. **Single realization**: No bootstrap or cross-validation uncertainty for real networks.

4. **Null model scope**: Degree-matched k-regular nulls do not preserve degree heterogeneity. Configuration model nulls are essential for networks with broad degree distributions.

5. **Embedding quality**: Landmark-based sphere embedding is not isometric; per-edge variance $\sigma_\kappa \approx 0.3$–$0.7$.

6. **Semi-analytical**: The Hehl-based framework yields an analytical $\eta_c(N, C)$ validated on random regular graphs (MAE $= 0.009$), but the matching heuristic is empirical, not proven. The analytical η_c overpredicts empirical values by ∼5–15%. No closed-form derivation of C^* exists.

7. **Domain shift**: The $C^* = 0.05$ model is trained on semantic networks with $C \in [0, 0.24]$ and fails on biological networks with $C \in [0.23, 0.88]$. A biological threshold $C^*_{\text{bio}} \approx 0.22$ is suggested but requires additional validation.

8. **Curvature notion**: Our analysis uses ORC exclusively. Forman–Ricci and LLY curvatures are always negative across all 16 networks (including Dutch SWOW), showing that the phase transition is specific to the ORC transport-geometric definition.

9. **Temporal dynamics**: All analyses are cross-sectional snapshots. Time-resolved ORC analysis [18] could reveal whether the phase boundary is crossed dynamically during network growth.

8 Code and Data Availability

All code, data, and results are available at:

https://github.com/agourakis82/hyperbolic-semantic-networks

Julia scripts: `unified_semantic_orc.jl`, `bridge_analysis.jl`, `hypercomplex_semantic_orc.jl`.
Sounio cross-validation: `experiments/06_hypercomplex/`. Lean 4 formalization: `lean/HyperbolicSemanticNe`
Results: `results/unified/`, `results/experiments/hypercomplex_sounio_*.csv`.

Table 9: Supplementary Table S1: ORC on SWOW-EN ($N = 438$, $E = 640$) for five values of the idleness parameter α. The sign of $\bar{\kappa}$ is invariant for $\alpha \in [0, 0.75]$; $\alpha = 1$ is degenerate (all mass on self, $\kappa \equiv 0$ identically). Standard choice $\alpha = 0.5$ (Ollivier 2009) lies at the center of the meaningful range.

α	$\bar{\kappa}$	σ_κ	SE	Geometry
0.00	−0.308	0.553	0.0218	Hyperbolic
0.25	−0.214	0.452	0.0179	Hyperbolic
0.50	−0.137	0.312	0.0124	Hyperbolic
0.75	−0.069	0.156	0.0062	Hyperbolic
1.00	0.000	0.000	0.0000	Degenerate †

† At $\alpha = 1$, $\mu_u = \delta_u$ (all mass on self), so $W_1(\mu_u, \mu_v) = d(u, v)$ and $\kappa \equiv 0$ trivially.

A Alpha Sensitivity Analysis

B Hypergraph ORC for Semantic Networks

Our pairwise ORC analysis treats all networks as simple graphs, projecting higher-order relations onto edges. This projection is a structural approximation for ConceptNet, WordNet, and BabelNet, whose fundamental objects are *relational triples* or *synsets*.

The projection loss. A ConceptNet triple such as (`dog`, `IsA`, `animal`) becomes a single edge {`dog`, `animal`}, discarding the relation type. WordNet synsets collapse to single nodes, making within-synset cohesion invisible to ORC. This projection systematically *underestimates* higher-order connectivity and explains why both networks appear near-Euclidean ($\bar{\kappa} \approx 0$) despite rich internal structure.

ORCHID framework. The ORCHID framework [25] generalises ORC to hypergraphs via hyperedge-weighted random walks: for a hyperedge $e = \{v_1, \ldots, v_m\}$, the transport plan distributes probability mass uniformly over e, and the Wasserstein-1 distance between neighbouring hyperedge distributions replaces the pairwise optimal transport.

Experimental results: a striking curvature reversal. We applied ORCHID to ConceptNet EN ($n_{\text{hyp}} = 305$ valid hyperedges) and WordNet EN ($n_{\text{hyp}} = 3088$ synset hyperedges).

Table 10: ORCHID hyperedge ORC vs. pairwise clique-expansion ORC. $\Delta\bar{\kappa} = \bar{\kappa}_{\text{ORCHID}} - \bar{\kappa}_{\text{pair}}$. The fraction negative is computed over all processed hyperedges. [EMPIRICAL]

Network	n_{hyp}	$\bar{\kappa}_{\text{ORCHID}}$	$\bar{\kappa}_{\text{pair}}$	$\Delta\bar{\kappa}$	Interpretation
WordNet EN	3088	+0.534	−0.101	**+0.635**	Synsets spherical; tree edges hyperbolic
ConceptNet EN	305	+0.504	+0.389	+0.116	Hyperedges more spherical than pairs

The WordNet result is the most striking: at the hyperedge (synset) level, curvature is *strongly positive* ($\bar{\kappa}_{\text{ORCHID}} = +0.534$), while pairwise projection is *negative* ($\bar{\kappa}_{\text{pair}} = -0.101$)—a reversal of $\Delta\bar{\kappa} = +0.635$. WordNet is simultaneously *spherical within synsets* and *hyperbolic between synsets*. The near-Euclidean aggregate $\bar{\kappa} \approx 0$ is a cancellation artefact from mixing two distinct geometric scales—the first empirical demonstration of *multi-scale geometric phase coexistence* in a semantic network. [EMPIRICAL]

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